



# Transportation Services

February 27, 2019



**BRECKSVILLE-BROADVIEW HTS.  
CITY SCHOOL DISTRICT**

*"where fine education is a heritage"*

## Routing Dynamics :

- Capacity - Seats available. Manufacturer's rating and the reality of being able to place two students in a seat.
  - 65-passenger rated bus holds 43 students, two students per seat
  - 72-passenger rated bus holds 48 students, two students per seat
  - 84-passenger rated bus holds 56 students, two students per seat
- Time – Available time for the bus to reach capacity. Is there enough time for the bus to pick up students and reach capacity? Three things commonly limit available time:
  - School times that are too close together.
  - Too many bus stops are on a route.
  - Distance the bus travels to fulfill route responsibilities.
- Fleet – The number of buses used to support pupil transportation. The availability of bus drivers adversely impacts the number of buses on the road.

# Transportation

## School Bus Fleet Comparison

| FY   | DISTRICT NAME                             | COUNTY          | Reg Ed<br>Bus Count | Square<br>Miles | Total<br>Riders > 1<br>Mile | Riders per<br>Bus | Riders per<br>Sq Mile |
|------|-------------------------------------------|-----------------|---------------------|-----------------|-----------------------------|-------------------|-----------------------|
| 2017 | <b>Brecksville-Broadview Heights City</b> | <b>Cuyahoga</b> | <b>36</b>           | <b>29</b>       | <b>2,093</b>                | <b>58</b>         | <b>72</b>             |
| 2017 | Bedford City                              | Cuyahoga        | 51                  | 20              | 1,633                       | 32                | 82                    |
| 2017 | Mayfield City                             | Cuyahoga        | 48                  | 22              | 1,703                       | 35                | 77                    |
| 2017 | North Royalton City                       | Cuyahoga        | 50                  | 25              | 2,650                       | 53                | 106                   |
| 2017 | Solon City                                | Cuyahoga        | 56                  | 23              | 2,239                       | 40                | 97                    |
| 2017 | Westlake City                             | Cuyahoga        | 42                  | 16              | 1,489                       | 35                | 93                    |
| 2017 | Hudson City (contract service)            | Summit          | 54                  | 30              | 2,517                       | 47                | 84                    |
| 2017 | Nordonia Hills City (contract service)    | Summit          | 36                  | 28              | 2,371                       | 66                | 85                    |

# Transportation

## N. Royalton Current Bell Schedules (3 Tier / 50 Buses)

| AM Bell Times      |             | PM Bell Times      |             |                              |
|--------------------|-------------|--------------------|-------------|------------------------------|
| <u>School</u>      | <u>Time</u> | <u>School</u>      | <u>Time</u> | <u>Length of Student Day</u> |
| High School        | 7:20 AM     | High School        | 2:22 PM     | 7 hours, 2 minutes           |
| Middle School      | 8:00 AM     | Middle School      | 3:00 PM     | 7 hours                      |
| Elementary Schools | 9:00 AM     | Elementary Schools | 3:30 PM     | 6 hours, 30 minutes          |

## Solon Current Bell Schedules (Hybrid 2 Tier / 56 Buses)

| AM Bell Times       |             | PM Bell Times      |             |                              |
|---------------------|-------------|--------------------|-------------|------------------------------|
| <u>School</u>       | <u>Time</u> | <u>School</u>      | <u>Time</u> | <u>Length of Student Day</u> |
| High School         | 7:50 AM     | High School        | 2:55 PM     | 7 hours, 5 minutes           |
| Elementary Schools  | 8:00 AM     | Elementary Schools | 2:30 PM     | 6 hours, 30 minutes          |
| Middle School 7 & 8 | 8:40 AM     | Middle School      | 3:30 PM     | 6 hours, 50 minutes          |
| Middle School 5 & 6 | 8:50 AM     | Middle School      | 3:40 PM     | 6 hours, 50 minutes          |

## Nordonia Hills Current Bell Schedules (3 Tier / 36 Buses)

| AM Bell Times      |             | PM Bell Times      |             |                              |
|--------------------|-------------|--------------------|-------------|------------------------------|
| <u>School</u>      | <u>Time</u> | <u>School</u>      | <u>Time</u> | <u>Length of Student Day</u> |
| High School        | 7:20 AM     | High School        | 2:11 PM     | 6 hours, 51 minutes          |
| Middle School      | 8:00 AM     | Middle School      | 2:45 PM     | 6 hours, 45 minutes          |
| Elementary Schools | 9:00 AM     | Elementary Schools | 3:30 PM     | 6 hours, 30 minutes          |

# Transportation

## BBHCSD Current Bell Schedules ( Hybrid 4 Tier / 36 Buses)

| AM Bell Times      |             | PM Bell Times      |             |                              |
|--------------------|-------------|--------------------|-------------|------------------------------|
| <u>School</u>      | <u>Time</u> | <u>School</u>      | <u>Time</u> | <u>Length of Student Day</u> |
| High School        | 7:20 AM     | High School        | 2:20 PM     | 7 hours                      |
| Middle School      | 7:50 AM     | Middle School      | 2:50 PM     | 7 hours                      |
| Central School     | 8:45 AM     | Central School     | 3:10 PM     | 6 hours, 25 minutes          |
| Elementary Schools | 9:25 AM     | Elementary Schools | 3:40 PM     | 6 hours, 15 minutes          |

- ❖ High School students are dropped off as early as 6:55 AM.
- ❖ Central School has buses unload students early to support the completion of the morning elementary school routes.
- ❖ PM High School buses are delayed 10 to 15 minutes daily exiting the site.
- ❖ PM Middle School buses pick up as late as 3:00 PM with Central School responsibilities.
- ❖ Central School buses pick up students as late as 3:30 PM with Elementary School responsibilities.
- ❖ Elementary Schools have buses that arrive as late as 3:55 to 4:00 PM regularly.

## Recommendations:

- ❖ Increase our fleet size by an ***additional*** five buses in advance of the 2019-2020 school year.
- ❖ Move forward with the two OSC bid buses scheduled for delivery in October increasing our Regular Ed fleet size by six during fiscal year 2019-2020. (a wheelchair lift bus is included in this purchase)
- ❖ Develop a substitute bus driver compensation model that supports an aggressive driver recruiting strategy.
- ❖ Commission a traffic study of school bus and automobile traffic patterns in and out of the High School and Middle School. Identify how we can improve the flow of additional buses through the site.
- ❖ Move to install GPS on all buses in advance of the 2019-2020 school year.

# Transportation

## Impact on Transportation Services Operating and Capital Expenses

| Fiscal Year 2018                                                                                                                | ODE Average Operating Cost Per Bus X 6 Regular Ed Buses  | BBHCSD Average Operating Cost Per Bus X 6 Regular Ed Buses |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|
| Approximate Annual Operating Cost Per Bus                                                                                       | \$65,000                                                 | \$55,000                                                   |
| Projected Operating Expense Impact                                                                                              | \$455,000                                                | \$385,000                                                  |
| Less Operating Expense Reduction From Central School Transportation Tier                                                        | \$170,000                                                | \$170,000                                                  |
| Range Of Annual Operating Expense Increase From The Addition Of 6 Regular Ed Buses To The Fleet                                 | <u>\$285,000</u>                                         | <u>\$215,000</u>                                           |
|                                                                                                                                 |                                                          |                                                            |
| School Bus Procurement Costs                                                                                                    | Ohio Schools Council Estimated Bid Price                 | Lease Purchase Option With 5 Year Term                     |
| January 2019 Authorization to Purchase One 84-Passenger Regular Ed Bus & One 54-Passenger Lift Bus For Delivery In October 2019 | \$193,500                                                | Not Applicable                                             |
| Five Addition Regular Ed Buses, Four 65-Passenger Buses & One 84-Passenger Bus, For Delivery Prior To The 2019-2020 School Year | \$412,000                                                | \$88,500 per year                                          |
|                                                                                                                                 |                                                          |                                                            |
| GPS                                                                                                                             | Projected Installation Cost & Firt Year Recuring Expense | Annual Recurring Expense                                   |
| Installation on 50 School Buses                                                                                                 | \$30,675                                                 | \$20,100                                                   |